THE PRIVATE CYBER LAB HANDBOOK

LAB 02

Group Policy Management & Windows 10 Security Configuration

Windows Server 2019 • Active Directory • Group Policy • Windows 10

Practical Lab Report • Version 2.0 (expanded and explained) • Based on observed session results

Domain: zroots.local
Domain Controller: DC01 (10.1.1.10)
Client: WIN10-01 (Windows 10)
GPO: Windows 10 Security Policy
Date: 12 October 2026

Document Control

Field	Details
Document title	Lab 02 Group Policy Management & Windows 10 Security Configuration
Version	2.0 expanded edition of the original v1.0 report (same evidence, fuller explanations)
Environment	Private Windows Server 2019 / Windows 10 lab
Domain	zroots.local (NetBIOS: ZROOTS)
Domain controller	DC01 10.1.1.10
Domain client	WIN10-01 Windows 10 build 10.0.19045.3803
Primary GPO	Windows 10 Security Policy
Evidence basis	Figures and command/report output supplied during the lab session
Status	GPO linking and computer-side application verified; scope caveat documented

How to Read This Report

Every section follows the same pattern: first **what was observed** in the lab, then **why it matters**, then any **caveat** that stops a reader from drawing a wrong conclusion. Observed facts come only from the supplied Figures and pasted command output. Explanatory text (definitions, rationale, recommendations) is general Windows/Active Directory knowledge added to help the reader understand the results; it is labelled as such where the difference matters.

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1. Executive Summary

This lab documented Group Policy Management for the Active Directory domain **zroots.local**. A Group Policy Object (GPO) named **"Windows 10 Security Policy"** was linked to the Windows 10 computer Organizational Unit (OU), and the domain-joined workstation **WIN10-01** was verified as receiving that GPO together with the **Default Domain Policy**. The client reached DC01 at 10.1.1.10, a Group Policy refresh completed, and the computer-scope Group Policy Results report showed no processing errors.

In plain terms: an administrator defines security rules once on the server, and Windows 10 machines placed in the right OU download and enforce those rules automatically. This lab proves that chain works end to end the policy exists, it is linked to the right place, the client fetched it, and the client says it applied it without errors.

The report also draws a deliberate line between a GPO being **applied** and every setting inside it being **appropriate for every account type**. Password and account-lockout settings placed in a workstation-linked GPO should not be assumed to govern domain user accounts (see Sections 8.1 and 13).

Overall outcome

Area	Observed outcome	What it means
Domain connectivity	PASS ping to 10.1.1.10 received 4/4 replies, 0% loss.	The client can reach the DC at the network level, a prerequisite for fetching policy.
Group Policy refresh	PASS computer and user policy updates completed successfully.	The client successfully contacted the DC and re-evaluated its policy.
Applied computer GPOs	PASS Windows 10 Security Policy and Default Domain Policy listed as applied.	Both expected GPOs are in scope and were processed for WIN10-01.
Computer-side processing	PASS No Errors Detected; Group Policy Infrastructure, Registry and Security statuses were Success.	The Group Policy engine and the relevant client-side extensions ran cleanly.
Security settings report	Configured password, lockout, audit and selected security options appeared with winning GPOs.	Settings can be traced back to the GPO that supplied them.
Policy-scope validation	Verify domain account password/lockout policy separately before claiming it applies to domain users.	Prevents a common misunderstanding about where account policy takes effect.

2. Objectives and Learning Outcomes

The lab had seven objectives. Next to each is the practical skill it builds and where it is evidenced in this report.

Objective	Why it matters	Evidence
Locate Group Policy Management and the domain/OU structure.	You cannot manage policy without knowing where objects live; OU design determines who receives which GPO.	Section 7; Figures B.3, B.6
Inspect the Windows 10 Security Policy GPO and its link to the intended OU.	A GPO does nothing until it is linked; the link is what points settings at machines.	Section 7; Figure B.11
Understand GPO status and computer vs. user configuration.	Each GPO has two halves; disabling an unused half reduces processing time and clarifies intent.	Section 7
Inspect security categories without making undocumented changes.	Reviewing before changing is safe practice and keeps the lab reproducible.	Section 8; Figures B.13–B.14
Refresh policy on WIN10-01 and verify applied GPOs with gpresult.	Verification on the client is the only proof that policy actually reached the target.	Section 9
Generate and review a computer-scope HTML Group Policy Results report.	The HTML report is a shareable, auditable record of what was applied and by which GPO.	Section 9.4
Record evidence and distinguish GPO application from policy scope.	Good documentation separates “it applied” from “it governs what I think it governs”.	Sections 12 and 13

3. Scope, Safety and Rules of Engagement

- **Authorized, isolated lab.** All work took place in a private lab. Review every setting before applying it to production; a lab-only value (for example a short lockout duration) can cause outages or lock out real users in a live domain.
- **Inspection only at the end.** The final verification steps (ping, gpupdate, gpresult, HTML report) only read and report. No new security settings were changed during those steps.
- **C:\GPO_Structure is documentation, not configuration.** It was a reference folder tree for organizing notes. Creating folders does not configure Group Policy; real settings are made only in the Group Policy Management Editor.
- **“Not Defined” does not mean “off”.** It means that specific GPO does not configure the item. Another GPO, or the Windows default, may still decide the behavior.
- **Respect scope.** Do not treat a workstation-linked GPO as authoritative for domain account password or lockout requirements.

4. Lab Environment

Component	Observed configuration	Role
Domain	zroots.local / ZROOTS	Active Directory domain the security boundary and policy container

Component	Observed configuration	Role
DC01	Windows Server 2019; 10.1.1.10	Domain controller; hosts Active Directory and the SYSVOL share that serves policy files
WIN10-01	Windows 10 build 10.0.19045.3803	Domain-joined client that receives and applies the GPO
Computer OU	OU=Windows 10,OU=Computer Accounts,DC=zroots,DC=local	Target OU for workstation GPOs; WIN10-01 sits here
Primary GPO	Windows 10 Security Policy	Computer-side security configuration
Other applied GPO	Default Domain Policy	Domain-linked baseline policy

OU layout observed

The Group Policy Management console shows a deliberate OU design under zroots.local: **Computer Accounts** (with child OUs **Windows 10** and **Windows 11**), **Domain Controllers**, **Groups**, **Servers**, **Service Accounts**, and **User Accounts** (with child OUs Administrators, Staff and Students). Separating Windows 10 and Windows 11 machines into their own OUs lets each operating system receive its own GPO without affecting the other. Separating users by role similarly allows role-specific user policy later.

5. Technical Background

Term	Meaning in this lab	Why it matters
GPO	Group Policy Object; a collection of settings linked to a site, domain or OU.	The unit of policy you create, link, enable and troubleshoot.
GPMC	Group Policy Management Console, used to create, link and inspect GPOs.	The main administrative console on DC01.
GPME	Group Policy Management Editor; opens when you Edit a GPO.	Where individual settings are actually changed.
Winning GPO	The GPO that Group Policy Results identifies as supplying a reported setting.	Lets you trace precedence when several GPOs touch the same setting.
gpupdate /force	Requests a refresh of computer and user Group Policy.	Avoids waiting for the periodic background refresh.
gpresult	Displays Resultant Set of Policy (RSoP) information for the selected scope.	The client-side proof of what was applied.
Computer scope	Policy processed for the computer account at startup/refresh; used for this report.	Applies regardless of who logs on.
User scope	Policy processed for a user account at logon/refresh; separate from computer-scope results.	Needs a user who has logged on to have data.
SYSVOL	Replicated folder on domain controllers that stores GPO files (templates, scripts).	AD holds GPO metadata; SYSVOL holds the settings files both must agree.

6. How Group Policy Is Processed

This section is background knowledge that explains why the results in this lab look the way they do.

6.1 Order of application (LSDOU)

Group Policy settings are applied in a fixed order. Where two GPOs configure the same setting, the one applied **later** wins:

1. **L**ocal — the local policy on the computer itself.
2. **S**ite — GPOs linked to the Active Directory site.
3. **D**omain — GPOs linked to the domain (here, Default Domain Policy).
4. **O**U — GPOs linked to OUs, from the outermost OU down to the OU that holds the computer or user (here, Windows 10 Security Policy on the Windows 10 OU).

In this lab, WIN10-01 receives the domain-level Default Domain Policy first, then the OU-level Windows 10 Security Policy. That is why the report credits Windows 10 Security Policy for the password, lockout and audit settings it defines, and credits Default Domain Policy only for the Security Options that Windows 10 Security Policy leaves undefined.

6.2 What makes a GPO "apply"

Condition	Check in this lab
The GPO exists and is enabled.	GPO Status: Enabled (Section 7).
The GPO is linked to a container above the target (site, domain or OU).	Linked to the Windows 10 OU; Link Enabled: Yes (Figure B.11).
The target object is inside that container.	WIN10-01 is in OU=Windows 10 (Figure B.9).
The client can reach the DC and read SYSVOL.	Ping succeeded; gpupdate completed.
Security filtering / WMI filtering does not exclude the target.	WMI Filter: None; GPO appeared under Applied GPOs.

6.3 Computer vs. user configuration

A GPO has two independent halves. **Computer Configuration** is applied to the machine at startup and on refresh; **User Configuration** is applied to the signed-in user at logon and on refresh. In this lab the GPO's User version is 0, meaning no user settings have ever been saved to it — all of its work is on the computer side. That is why verification used the computer scope.

7. GPO Creation, Linking and OU Placement

7.1 What was observed

The management Figures show the zroots.local domain, the Computer Accounts container, the Windows 10 OU, and the linked GPO. The GPO is named **"Windows 10 Security Policy."** Its

Details tab reported **GPO Status: Enabled**, **User version 0 (AD) / 0 (SYSVOL)**, and **Computer version 88 (AD) / 88 (SYSVOL)**.

The Linked Group Policy Objects tab for the Windows 10 OU shows the GPO at **Link Order 1**, **Enforced: No**, **Link Enabled: Yes**, **GPO Status: Enabled** and **WMI Filter: None** (Figure B.11). Before linking, the same tab was empty (Figure B.2).

7.2 Understanding the version numbers

Each GPO stores a version counter for its computer half and its user half, in two places: Active Directory (the GPO's metadata) and SYSVOL (the actual settings files). The counter rises each time a section is changed. Two things can be read from the lab values:

- **Matching AD and SYSVOL values (88 / 88)** indicate the two stores are in sync. A mismatch is a classic sign of replication trouble and can cause a client to apply stale or no settings.
- **User version 0** means no user configuration has ever been saved. The GPO is effectively computer-only.

7.3 GPO Status options

Status option	Effect	Use case
Enabled (selected)	Both computer and user sections are processed.	Normal operation.
User configuration settings disabled	Only the computer half is processed.	GPO contains only computer settings; skips user processing to speed logon.
Computer configuration settings disabled	Only the user half is processed.	GPO contains only user settings.
All settings disabled	Nothing is processed, but the GPO stays linked.	Temporarily switching a GPO off without unlinking it.

Enabled permits both sections to be processed; individual settings must still be configured to affect a system. A possible refinement is to set "User configuration settings disabled" on this GPO because its user version is 0 a small optimization, not a requirement.

7.4 Moving the computer into the right OU

Figure B.8 shows WIN10-01 and WINDOWS11 in the default **Computers** container, while Figure B.9 shows WIN10-01 inside the **Windows 10** OU. The computer account had to be placed in the OU, because a GPO linked to an OU only affects objects inside it (or in child OUs). The default Computers container is not an OU and cannot have a GPO linked to it, which is why a dedicated OU is used.

7.5 Editor areas inspected

- Computer Configuration → Policies → Windows Settings → Security Settings
- Account Policies: Password Policy, Account Lockout Policy and Kerberos Policy
- Local Policies: Audit Policy, User Rights Assignment and Security Options

- Event Log, Windows Defender Firewall with Advanced Security, Advanced Audit Policy Configuration
- Restricted Groups, System Services, Registry and File System

System Services startup options and permissions were Not Defined. Registry was empty. File System and Central Access Policy were empty. Windows Defender Firewall with Advanced Security had already been checked earlier in the session. Several other areas were empty or showed Not Defined entries. The practical conclusion is that this GPO's real content is concentrated in Account Policies, Audit Policy and a few security options.

8. Security Settings Inspected

The tables below come from the client-side Group Policy Results report. Each row shows the setting, its reported value, and the **winning GPO**. The third column in each table adds the purpose of the setting; it is explanatory and not part of the observed evidence.

8.1 Password Policy

Policy	Reported setting (winning GPO: Windows 10 Security Policy)	What it does
Enforce password history	5 passwords remembered	Stops users re-using their last 5 passwords.
Maximum password age	90 days	Forces a change at least every 90 days.
Minimum password age	1 day	Prevents cycling through changes to get back to an old password.
Minimum password length	12 characters	Longer passwords resist brute-force attacks far better than short ones.
Password complexity	Enabled	Requires a mix of character types and bans easy patterns such as the username.
Reversible encryption	Disabled	Passwords are not stored in a recoverable form the safe setting.

> **Important scope caveat**
> Account policies (password, lockout, Kerberos) work differently from other settings. On a domain, the policy that governs **domain user accounts** is taken from a GPO linked at the **domain root** (normally Default Domain Policy), not from a GPO linked to a workstation OU. A GPO linked to the Windows 10 OU can only influence the **local accounts on those workstations**. The report shows these values because the workstation processed them, not because they control domain logons.

8.2 Account Lockout Policy

Policy	Reported setting (winning GPO: Windows 10 Security Policy)	What it does
Account lockout duration	15 minutes	How long a locked account stays locked before auto-unlocking.
Account lockout threshold	5 invalid logon attempts	Failed attempts allowed before lockout slows password guessing.
Allow administrator account lockout	Enabled	Lets the built-in Administrator be locked out too, closing a brute-force gap.
Reset account lockout counter after	10 minutes	Time after which the failed-attempt counter returns to zero.

The three numeric values work together: five failed attempts inside a ten-minute window locks the account for fifteen minutes. The reset window must not be longer than the lockout duration, and this configuration respects that rule. The same scope caveat as Section 8.1 applies.

8.3 Audit Policy

Audit category	Reported setting	What it records
Audit account logon events	Success, Failure	Credential validation events for logons.
Audit account management	Success, Failure	Creating, changing or deleting users, groups and computers.
Audit directory service access	Success, Failure	Access to Active Directory objects (mainly relevant on DCs).
Audit logon events	Success, Failure	Interactive, network and service logons on the machine.
Audit object access	Failure	Failed attempts to open audited files, folders or registry keys.
Audit policy change	Success, Failure	Changes to audit, trust and user-right policies.
Audit privilege use	Failure	Failed use of sensitive privileges.
Audit process tracking	Success	Process creation and related activity.
Audit system events	Success, Failure	Startup, shutdown and security-subsystem events.

All nine are supplied by Windows 10 Security Policy. Auditing is the “camera” side of security: it does not block anything, but it creates the event-log trail used for investigations. Note that Audit object access and Audit directory service access only generate events for objects that also have a **System Access Control List (SACL)** entry, so enabling the category alone is not enough. Windows also offers **Advanced Audit Policy Configuration**, a finer-grained alternative; mixing it with the basic audit categories can cause conflicts unless the “Force audit policy subcategory settings” option is set deliberately.

8.4 Selected Security Options

Setting	Reported value / winning GPO	What it does
Allow anonymous SID/Name translation	Disabled Default Domain Policy	Stops unauthenticated users translating SIDs to account names, which limits account enumeration.
Do not store LAN Manager hash value on next password change	Enabled Default Domain Policy	Prevents storage of the weak legacy LM hash.
Force logoff when logon hours expire	Disabled Default Domain Policy	Users are not forcibly signed out when permitted hours end.

These three come from **Default Domain Policy**, not from Windows 10 Security Policy a live example of precedence: where the OU-level GPO is silent, the domain-level one still supplies the value. The report also showed **Certificate Services Client Auto-Enrollment Settings** with automatic certificate management shown as Enabled [Default setting], while the two displayed enrollment options were Disabled. No custom Registry, File System or System Services settings were found.

9. Client Refresh and Result Verification

All commands in this section were run on WIN10-01. They follow the logical order of a verification: can I reach the server, did I fetch policy, what was applied, and is there a record I can keep.

9.1 Connectivity

```
ping 10.1.1.10
```

Observed: 4 packets sent, 4 received, 0% loss; replies under 1 ms with TTL 128.
Interpretation: ICMP reachability is confirmed, so basic network routing between client and DC works. The very low latency and TTL 128 are consistent with a host on the same virtual network running Windows. Ping alone does not prove Group Policy will work (that also needs DNS and SMB access to SYSVOL), but a failure here would explain most policy failures.

9.2 Refresh policy

```
gpupdate /force
```

Observed: Computer Policy update and User Policy update both completed successfully.
Interpretation: By default clients re-check policy roughly every 90 minutes (with a random offset), so `/force` is used in labs to skip the wait and re-apply all settings. Some settings (software installation, folder redirection) may still need a logoff or restart to take effect.

9.3 Applied computer GPOs

```
gpresult /r /scope computer
```

Observed applied GPOs: Windows 10 Security Policy and Default Domain Policy. The computer DN was CN=WIN10-01,OU=Windows 10,OU=Computer Accounts,DC=zroots,DC=local. Policy was applied from DC01.zroots.local.

Interpretation: This confirms three things at once the client sits in the intended OU, it contacted the intended DC, and both expected GPOs passed filtering and were processed.

9.4 Computer HTML report

```
gpresult /scope computer /h C:\GPO-Computer-Report.html
```

Observed: The command completed without an error and the report opened in a browser. It showed "No Errors Detected"; Group Policy Infrastructure, Registry and Security component statuses were Success. Both expected GPOs appeared under Applied GPOs.

A repeated run returned "A file by that name already exists." This meant the file already existed **not** that policy processing failed. To regenerate it, add `/f`:

```
gpresult /scope computer /h C:\GPO-Computer-Report.html /f
```

9.5 User-scope report message

```
gpresult /h "%USERPROFILE%\Desktop\GPO-Report.html"
```

Observed message: INFO: The user "ZROOTS\Administrator" does not have RSoP data.

Interpretation: User-scope RSoP data exists only for a user who has actually logged on and had user policy processed on that machine. Because this GPO has no user settings (User version 0), the message is expected and does not negate the successful computer-scope report. The computer-only command is the correct tool for workstation verification.

10. Step-by-Step Reproduction Guide

A reader can repeat the lab with the steps below. They are written from the evidence in this report; adapt names to your own environment.

On DC01 (Windows Server 2019)

1. Open **Server Manager** → **Tools** → **Group Policy Management** (Figure B.4 shows the dashboard; Figure B.3 the console).
2. Expand **Forest: zroots.local** → **Domains** → **zroots.local** → **Computer Accounts** and select the **Windows 10** OU.
3. Confirm the OU is empty of linked GPOs on the **Linked Group Policy Objects** tab (Figure B.2).
4. Open **Active Directory Users and Computers** and move the computer account **WIN10-01** from the **Computers** container into **Computer Accounts** → **Windows 10** (Figures B.8, B.9).
5. Create or select the GPO **Windows 10 Security Policy** and link it to the Windows 10 OU (right-click the OU → Link an Existing GPO). Verify Link Enabled = Yes (Figure B.11).
6. Right-click the GPO → **Edit** to open the Group Policy Management Editor (Figure B.12).

7. Navigate **Computer Configuration → Policies → Windows Settings → Security Settings** and review or set Password, Lockout and Audit settings (Figures B.13, B.14).
8. On the GPO's **Details** tab, confirm Status is Enabled and that AD and SYSVOL version numbers match.

On WIN10-01 (Windows 10 client)

1. Run `ping 10.1.1.10` to confirm the DC is reachable.
2. Run `gpupdate /force` from an elevated Command Prompt and wait for both updates to succeed.
3. Run `gpresult /r /scope computer` and confirm both GPOs under Applied Group Policy Objects.
4. Run `gpresult /scope computer /h C:\GPO-Computer-Report.html` (add `/f` to overwrite), open the report and check for "No Errors Detected".

11. Troubleshooting Notes

Observation	Interpretation / action
Ping to DC01 succeeded	Basic ICMP reachability from WIN10-01 to 10.1.1.10 was confirmed. If it fails, check IP settings, DNS and the virtual network first.
gpresult lists both GPOs	Computer-side GPO application is confirmed.
User-scope RSoP data unavailable	User-scope data is separate and needs a logged-on user with user policy; use /scope computer for this workstation check.
HTML report already exists	Use /f to overwrite or open the existing report; this is a file-name message, not a policy failure.
Report summary says "No data available"	The summary header can be empty for a section while the detailed computer section still shows no errors, successful components and applied GPOs. Judge by the detailed section.
If a GPO were missing from Applied GPOs	Check, in order: computer is in the right OU, link is enabled, GPO status is enabled, security filtering and WMI filters, then AD/SYSVOL version match.
If a setting shows an unexpected winner	Open the HTML report's Winning GPO column and review link order and any Enforced links.

12. Results and Evidence Register

Control / task	Evidence	Result
Domain/OU context	GPMC and Active Directory Figures (B.3, B.5–B.9)	Documented
GPO link	Linked GPO Figure (B.11)	Verified
GPO editor navigation	Computer Configuration / Windows Settings Figures (B.12–B.14)	Documented

Control / task	Evidence	Result
Security policy values	Pasted HTML report output	Verified in report
Connectivity	Pasted ping output	PASS
Policy refresh	Pasted gpupdate output	PASS
Applied GPO list	gpresult output and HTML report text	PASS
HTML report	C:\GPO-Computer-Report.html on WIN10-01	Created/opened

Evidence note

Final ping, gpupdate and gpresult results and the HTML report content were supplied as pasted text. No separate Figure of those final command outputs was provided, and none has been fabricated. For a stricter submission, re-run the commands and capture Figures of the Command Prompt and of the report's "Applied GPOs" and "Component Status" sections.

13. Security Lessons and Scope Notes

- **Applied is not the same as complete.** A GPO under Applied GPOs was in scope and processed; that does not mean every possible setting is defined.
- **Not Defined means "this GPO does not set it".** Behavior then falls back to another GPO or the Windows default.
- **Winning GPO gives traceability.** Group Policy Results shows which GPO supplied each displayed setting, which makes precedence disputes solvable.
- **Account policy scope.** The workstation-linked GPO's Password and Lockout policies appeared in the workstation's computer results. Do not assume this establishes policy for domain user accounts; verify at the domain level (Default Domain Policy) or through Fine-Grained Password Policies.
- **Auditing needs more than a checkbox.** Audit settings may require SACLs or event-specific configuration before expected events are generated, and the event log must be large enough to retain them.
- **Documentation folders are not policy.** The C:\GPO_Structure tree is for organization only; real policy is set in the Group Policy Management Editor.
- **"No Errors Detected" has a narrow meaning.** It says the Group Policy engine ran without errors, not that the configuration is secure.

14. Recommendations and Next Steps

These suggestions go beyond the lab's observed scope and are offered as good practice. Check them against your organization's baseline (for example a CIS or Microsoft security baseline) before adopting.

Recommendation	Reason
Verify domain-level password and lockout policy in Default Domain Policy or a Fine-Grained Password Policy, and record the result.	Closes the scope gap identified in this report.
Review password values against current guidance.	Many benchmarks favor longer minimum lengths and longer password history than 12 and 5, and modern guidance questions mandatory frequent rotation. Treat the lab values as a starting point.
Consider moving to Advanced Audit Policy Configuration.	Finer-grained categories give more useful, less noisy logs than the nine basic categories.
Set “User configuration settings disabled” on Windows 10 Security Policy if it will remain computer-only.	Slightly faster logon and clearer intent.
Create a matching GPO for the Windows 11 OU.	The OU exists but no policy was demonstrated for it; keep both generations consistent.
Back up the GPO and export the HTML report to a safe location.	Allows rollback and gives an auditable before/after record.
Test every change on a pilot OU or machine first.	Prevents a bad setting reaching all workstations at once.

15. Knowledge Check

What does gpupdate /force do?

Answer: Requests a refresh of computer and user Group Policy, re-applying all settings instead of only changed ones.

Which command verifies computer-scope applied GPOs?

Answer: gpresult /r /scope computer.

Does “No Errors Detected” mean a complete security audit passed?

Answer: No. It means the report detected no Group Policy processing errors.

Does an applied GPO mean all settings are configured?

Answer: No. Individual settings may be Not Defined or use defaults.

Where should domain user password policy be verified?

Answer: At the domain level (Default Domain Policy) or through applicable Fine-Grained Password Policies.

Why did the report credit Default Domain Policy for three Security Options?

Answer: Windows 10 Security Policy does not define them, so the domain-level GPO remained the winning source.

What does a mismatch between AD and SYSVOL version numbers suggest?

Answer: A replication or sync problem, which can cause clients to apply stale or incomplete policy.

Why did the user-scope report say there was no RSoP data?

Answer: No user policy had been processed for that user on the machine; the GPO has no user settings.

16. Conclusion

The core Lab 02 objective managing the Windows 10 Security Policy GPO and verifying that WIN10-01 processes it was demonstrated by the linked-GPO evidence, the successful policy refresh, the gpresult output and the computer-scope HTML report. Both Windows 10 Security Policy and Default Domain Policy were listed as applied, with no processing errors and successful Group Policy Infrastructure, Registry and Security components.

Beyond the pass/fail result, the lab illustrates three durable lessons: policy only takes effect when GPO, link and target object line up; the winning-GPO view makes precedence traceable; and account-policy settings need to be verified at the correct scope.

Lab 02 is documented as completed for GPO linking and computer-side application verification. Domain-account password and lockout policy scope should be checked separately before claiming those settings govern domain users.

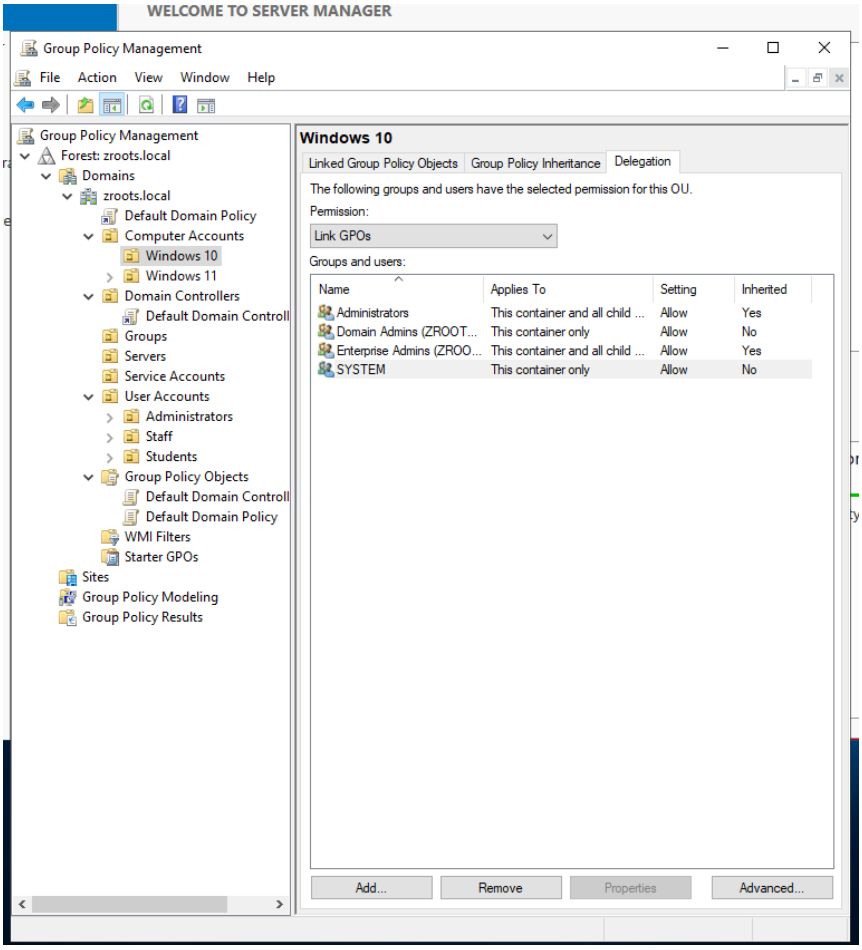
Appendix A Command Reference

Command	Purpose	Notes
ping 10.1.1.10	Check reachability to DC01.	Network-level test only.
gpupdate /force	Refresh computer and user Group Policy.	Some settings still need logoff or restart.
gpresult /r /scope computer	List computer-scope applied GPOs.	Run elevated for full computer data.
gpresult /h "%USERPROFILE%\Desktop\GPO-Report.html"	Generate a combined report.	May report missing user RSoP data.
gpresult /scope computer /h C:\GPO-Computer-Report.html	Generate a computer-scope HTML report.	Fails if the file already exists.
gpresult /scope computer /h C:\GPO-Computer-Report.html /f	Overwrite the existing computer report.	/f forces overwrite.

Appendix B Figure Evidence

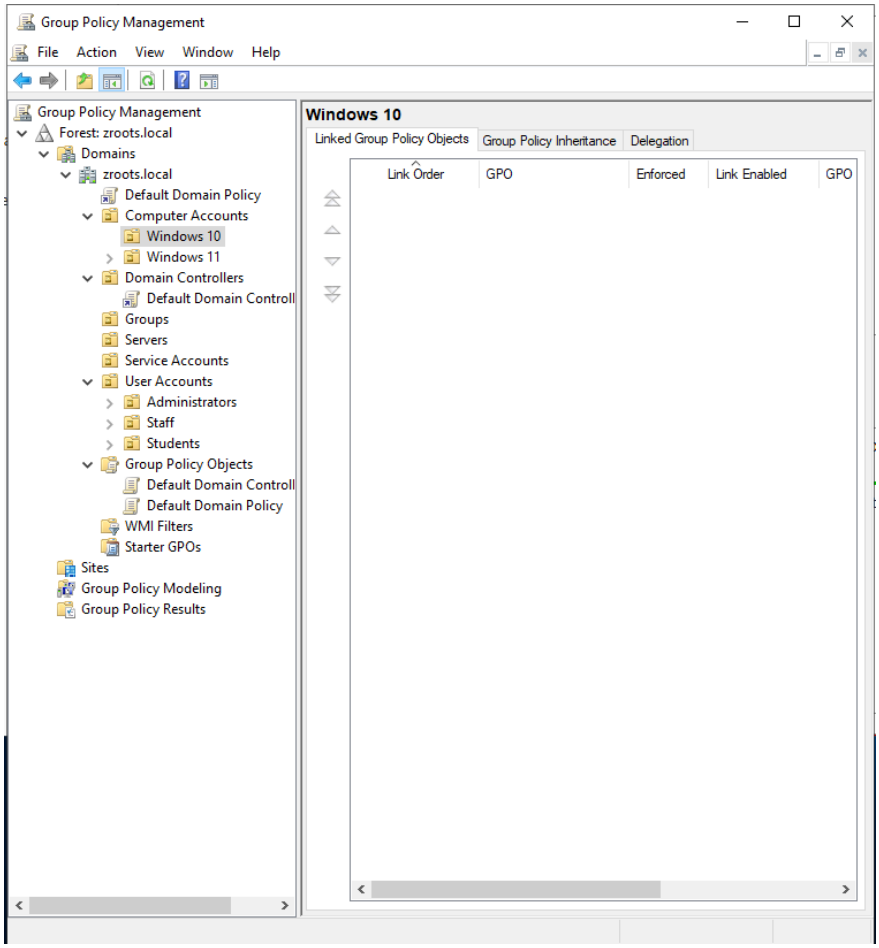
The Figures below were available from the Lab 02 session. Each figure is followed by a short explanation of what is visible and why it is relevant. Figures are not treated as proof of settings that are not visible in them.

Figure B.1 Windows 10 OU in Group Policy Management Delegation tab



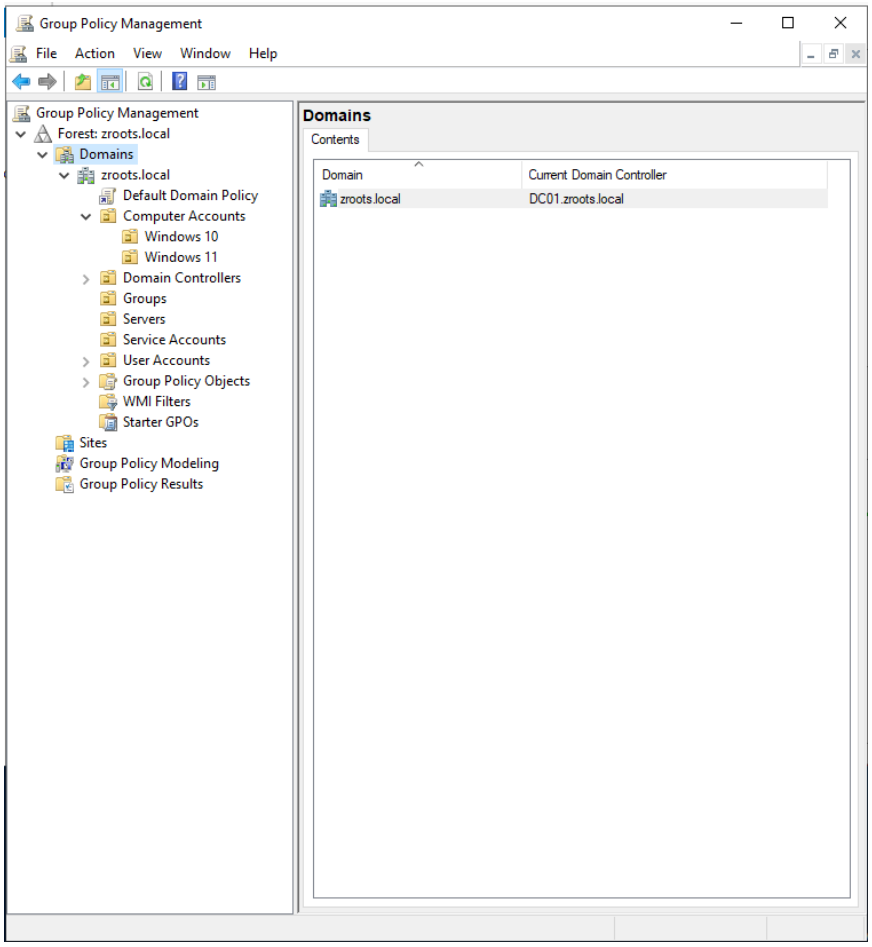
What it shows: *The Delegation tab for the Windows 10 OU with the "Link GPOs" permission selected. Administrators, Domain Admins, Enterprise Admins and SYSTEM are listed, showing who is allowed to link GPOs here. The tree on the left also reveals the OU structure.*

Figure B.2 Windows 10 OU linked Group Policy Objects before linking



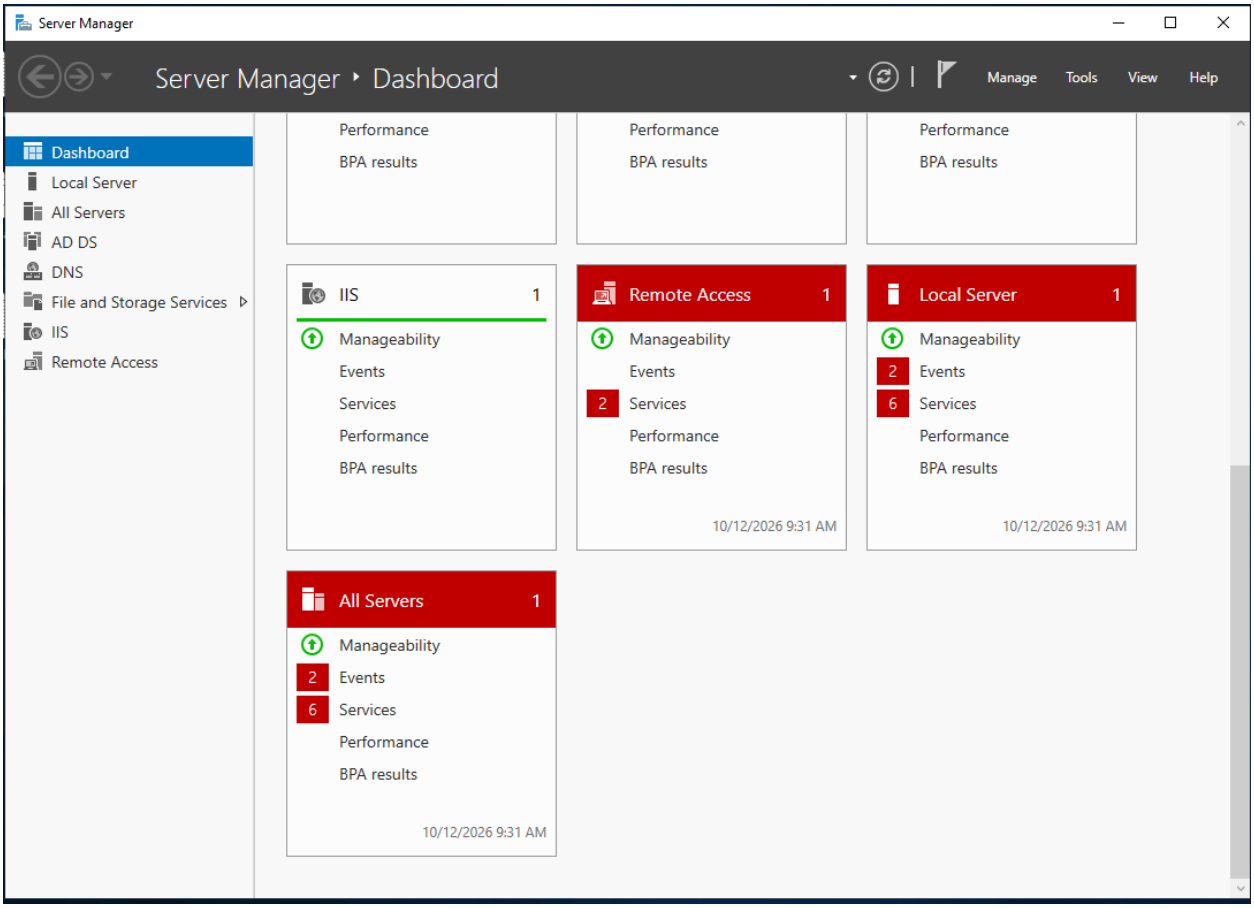
What it shows: *The Linked Group Policy Objects tab for the Windows 10 OU is empty, which is the "before" state prior to linking Windows 10 Security Policy.*

Figure B.3 Group Policy Management domain overview



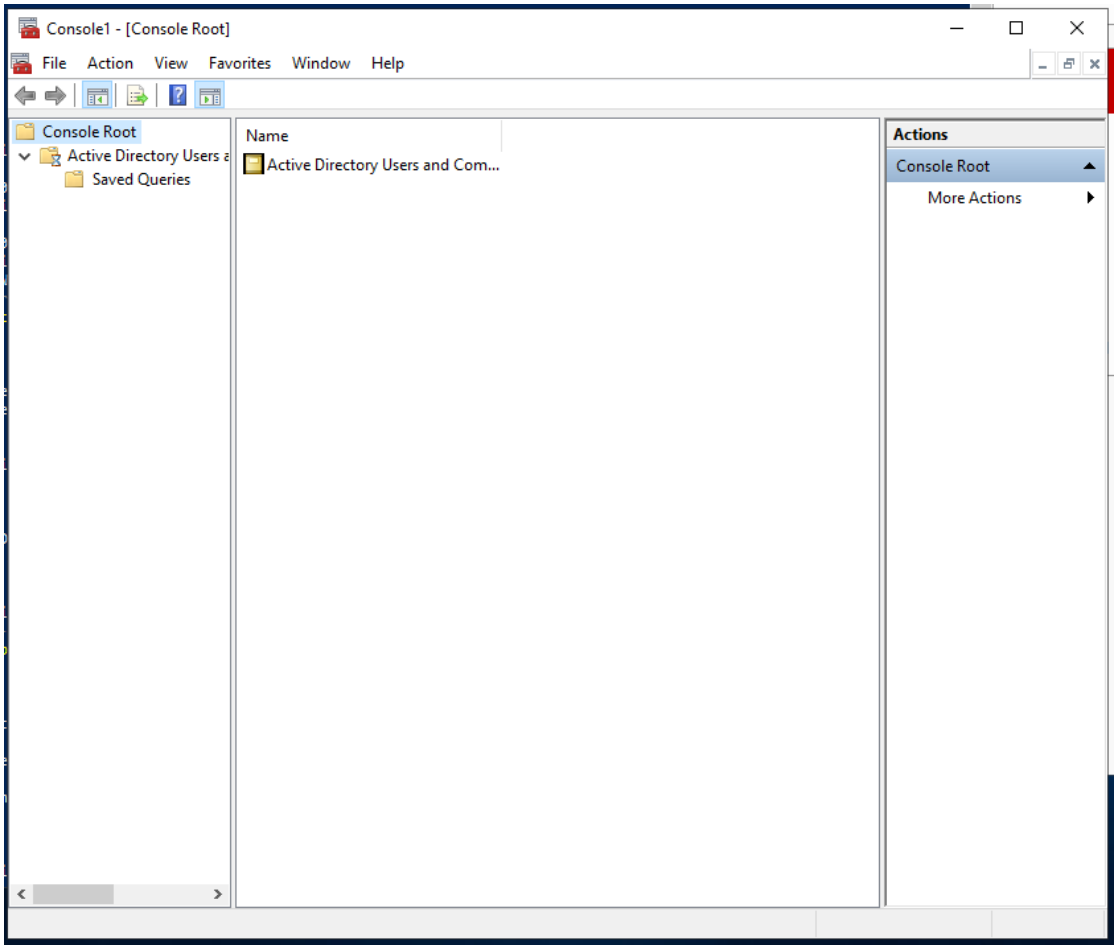
What it shows: *The zroots.local domain with DC01.zroots.local as its current domain controller, and the Default Domain Policy, Computer Accounts, Windows 10/11 OUs and Group Policy Objects nodes visible.*

Figure B.4 DC01 Server Manager dashboard



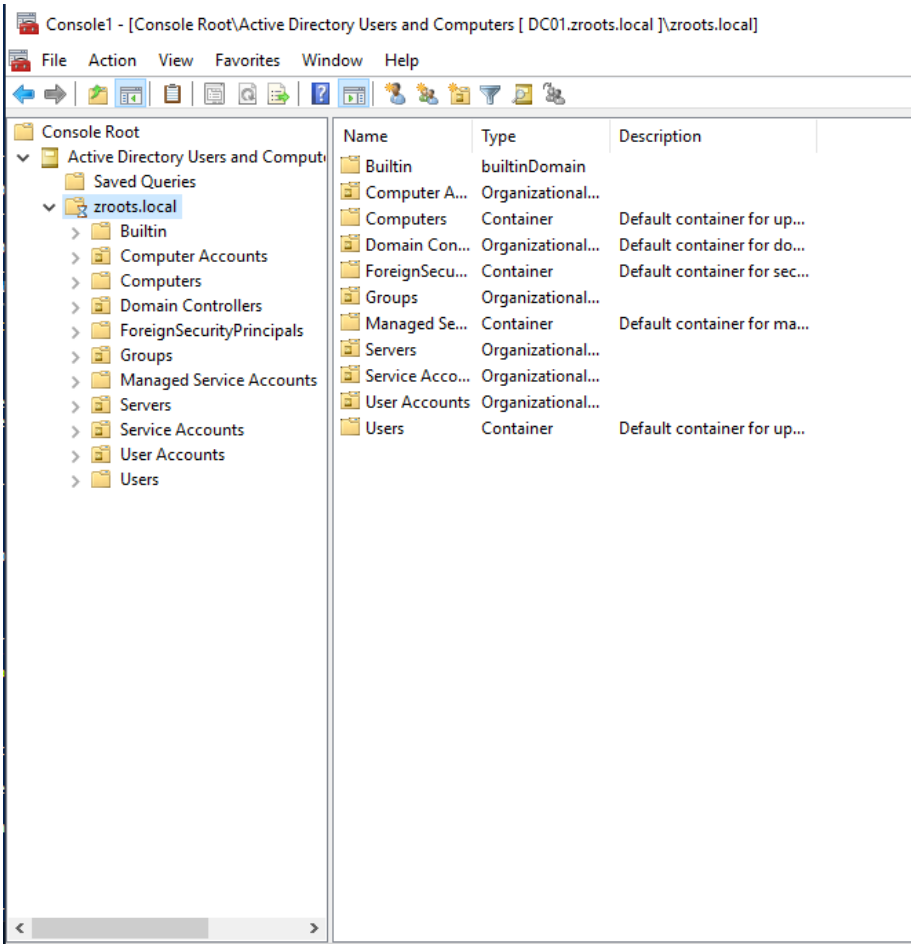
What it shows: *Server Manager on the domain controller, the starting point for opening Group Policy Management. The red tiles indicate services or events needing attention on this lab server.*

Figure B.5 Microsoft Management Console Active Directory Users and Computers snap-in



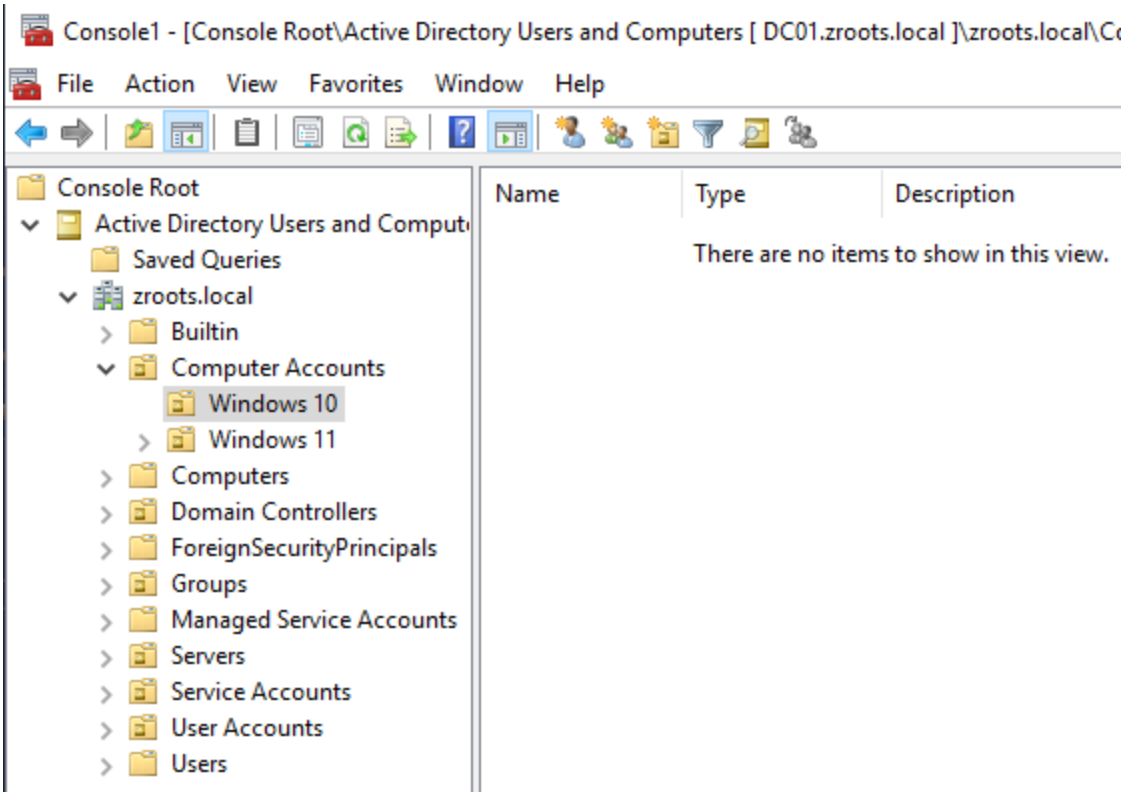
What it shows: *An empty MMC console with the Active Directory Users and Computers snap-in loaded, used to browse and move computer objects.*

Figure B.6 Active Directory domain containers and organizational units



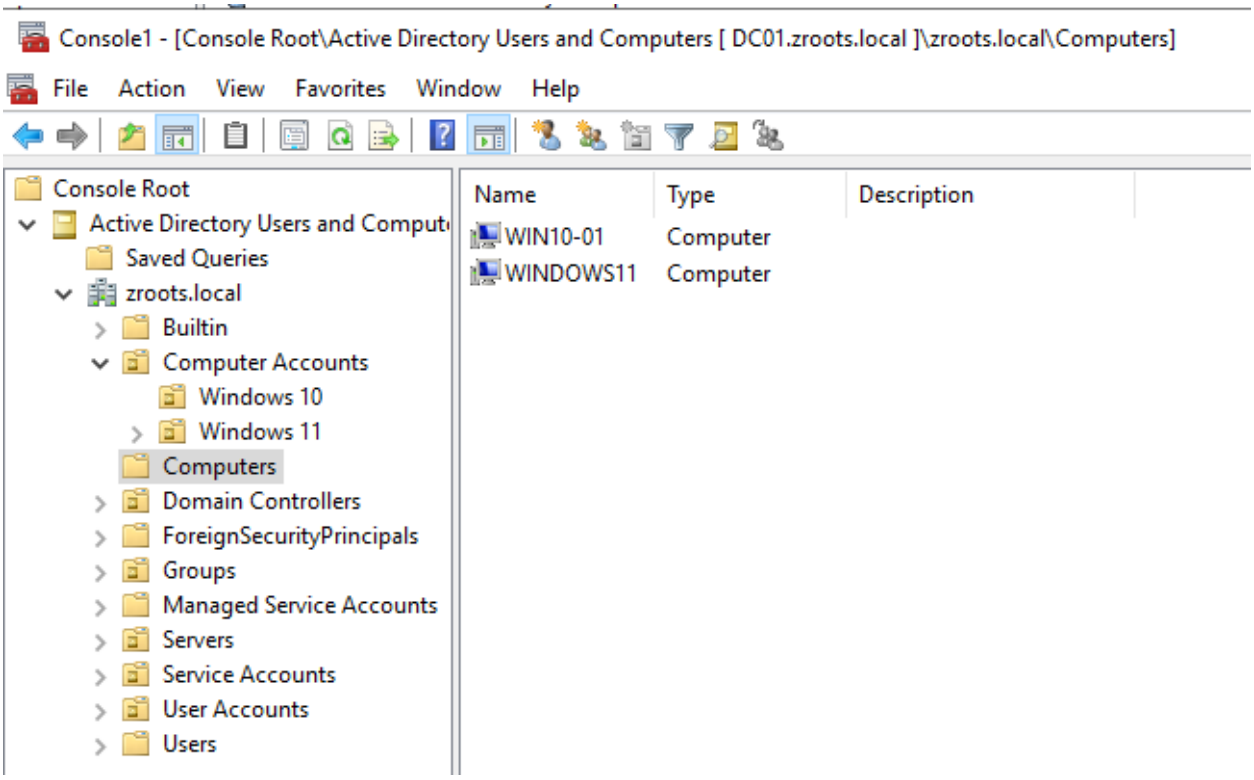
What it shows: *The zroots.local tree and contents: Builtin, Computers, Domain Controllers and the custom OUs (Computer Accounts, Groups, Servers, Service Accounts, User Accounts).*

Figure B.7 Windows 10 OU selected in Active Directory Users and Computers



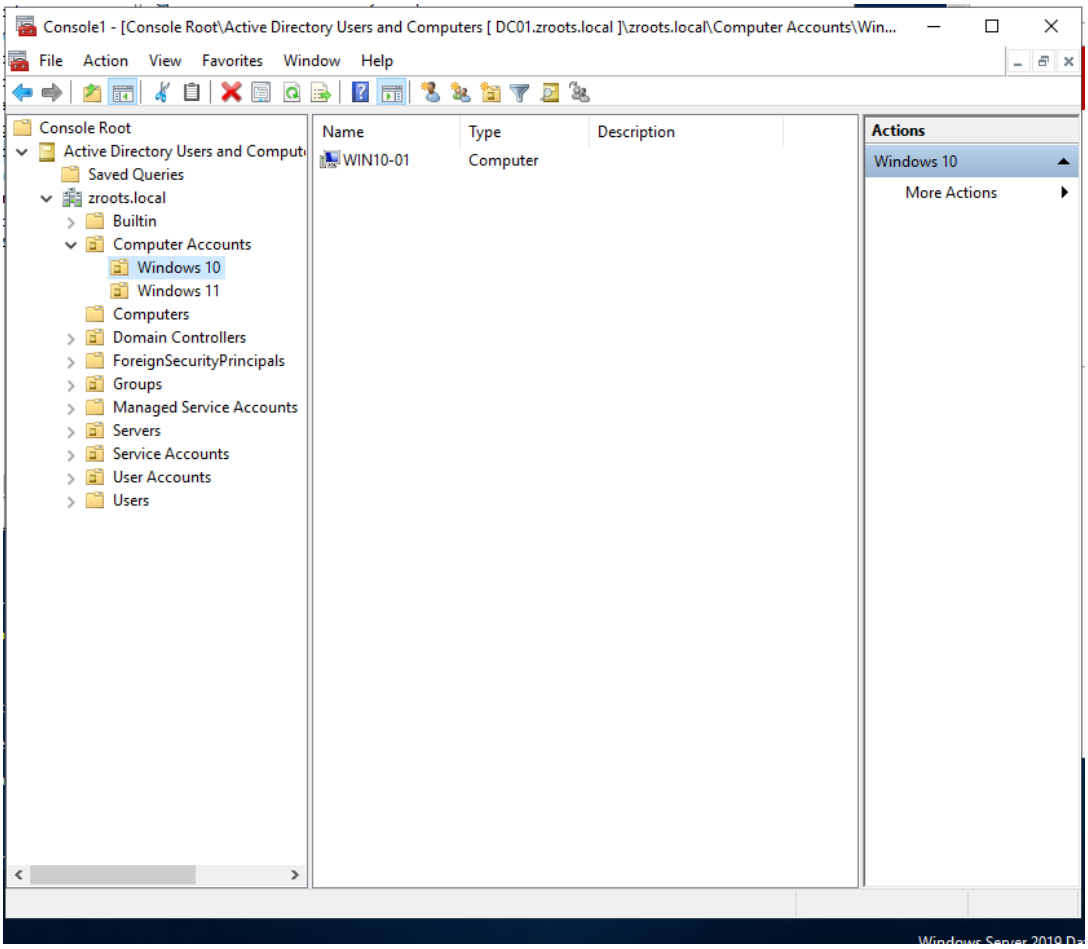
What it shows: *The Windows 10 OU selected under Computer Accounts, with "There are no items to show in this view" the OU was still empty at this point.*

Figure B.8 Computers container showing domain computer objects



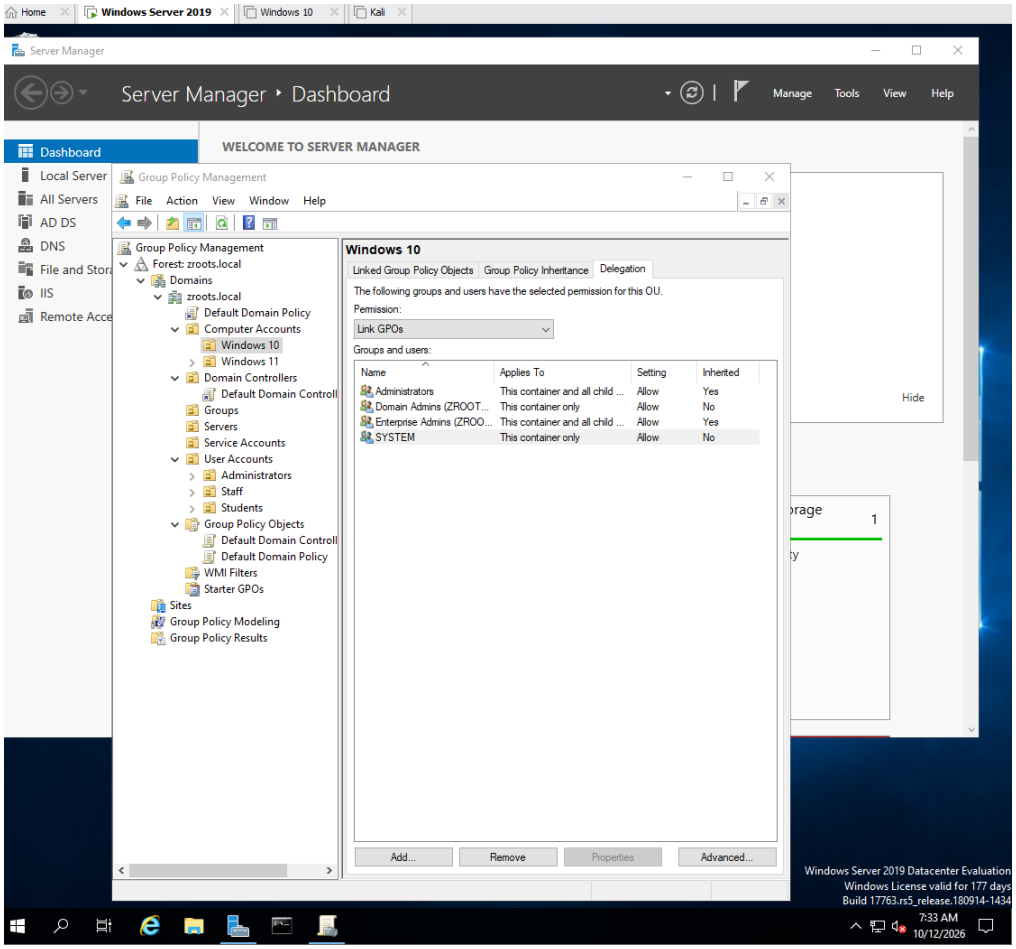
What it shows: *WIN10-01 and WINDOWS11 sitting in the default Computers container, which cannot receive linked GPOs.*

Figure B.9 Windows 10 OU showing WIN10-01



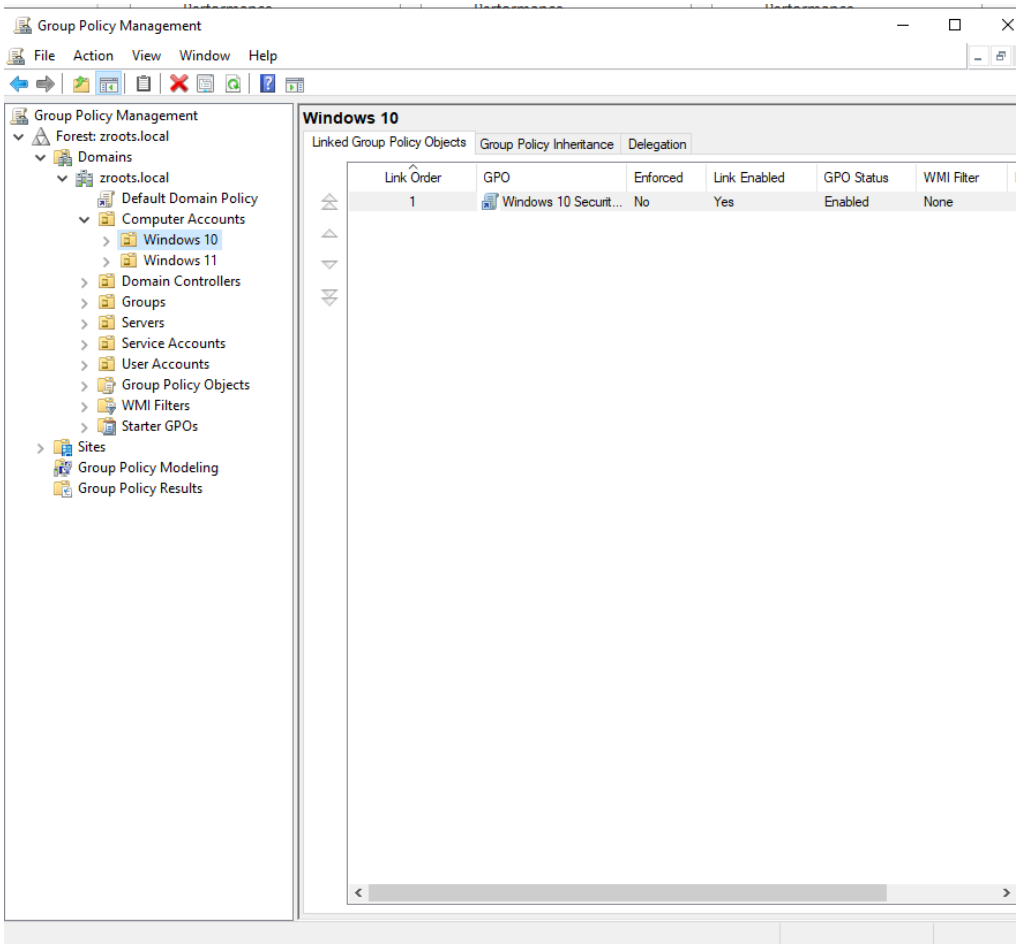
What it shows: *WIN10-01 now inside the Windows 10 OU, so a GPO linked there will reach it.*

Figure B.10 Group Policy Management Windows 10 OU delegation / link context



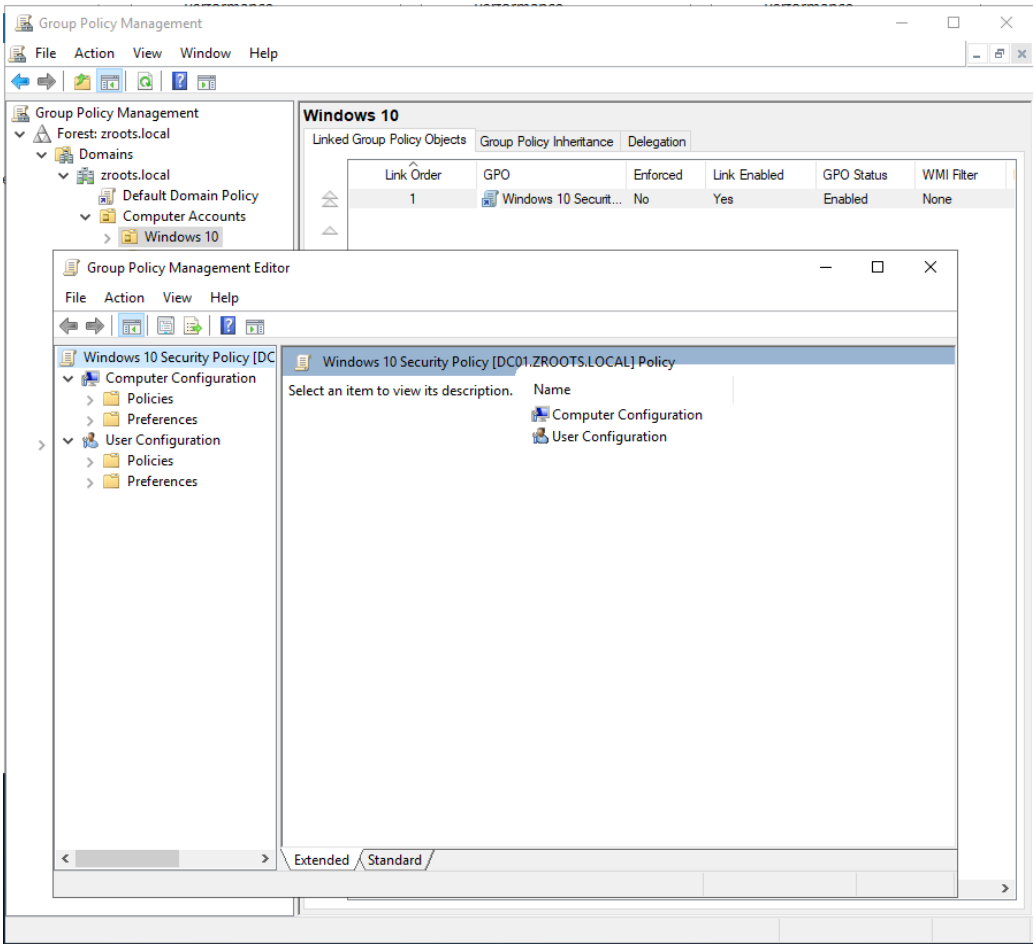
What it shows: *A wider view of the lab desktop with Group Policy Management open over Server Manager, showing the Windows 10 OU delegation view.*

Figure B.11 Windows 10 OU with Windows 10 Security Policy linked



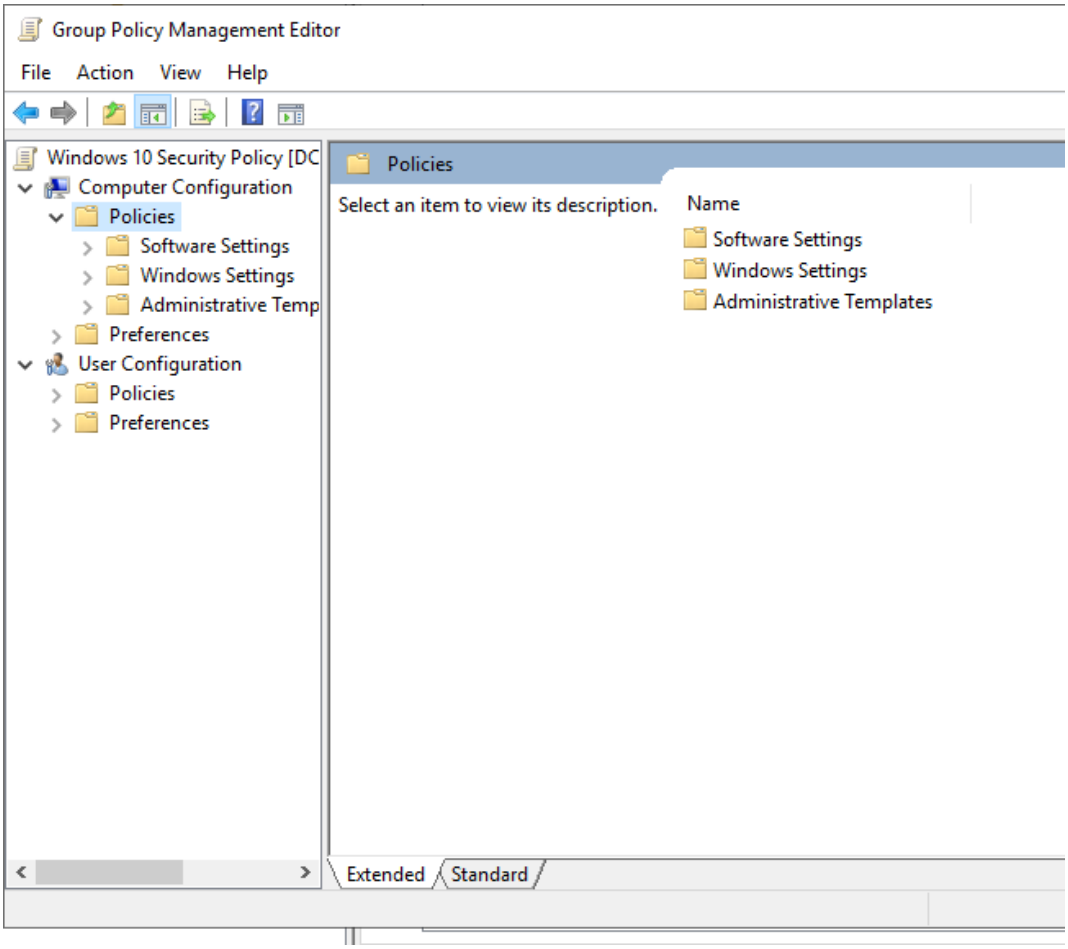
What it shows: *The key evidence of linking: Link Order 1, Enforced No, Link Enabled Yes, GPO Status Enabled, WMI Filter None.*

Figure B.12 GPO linked and Group Policy Management Editor open



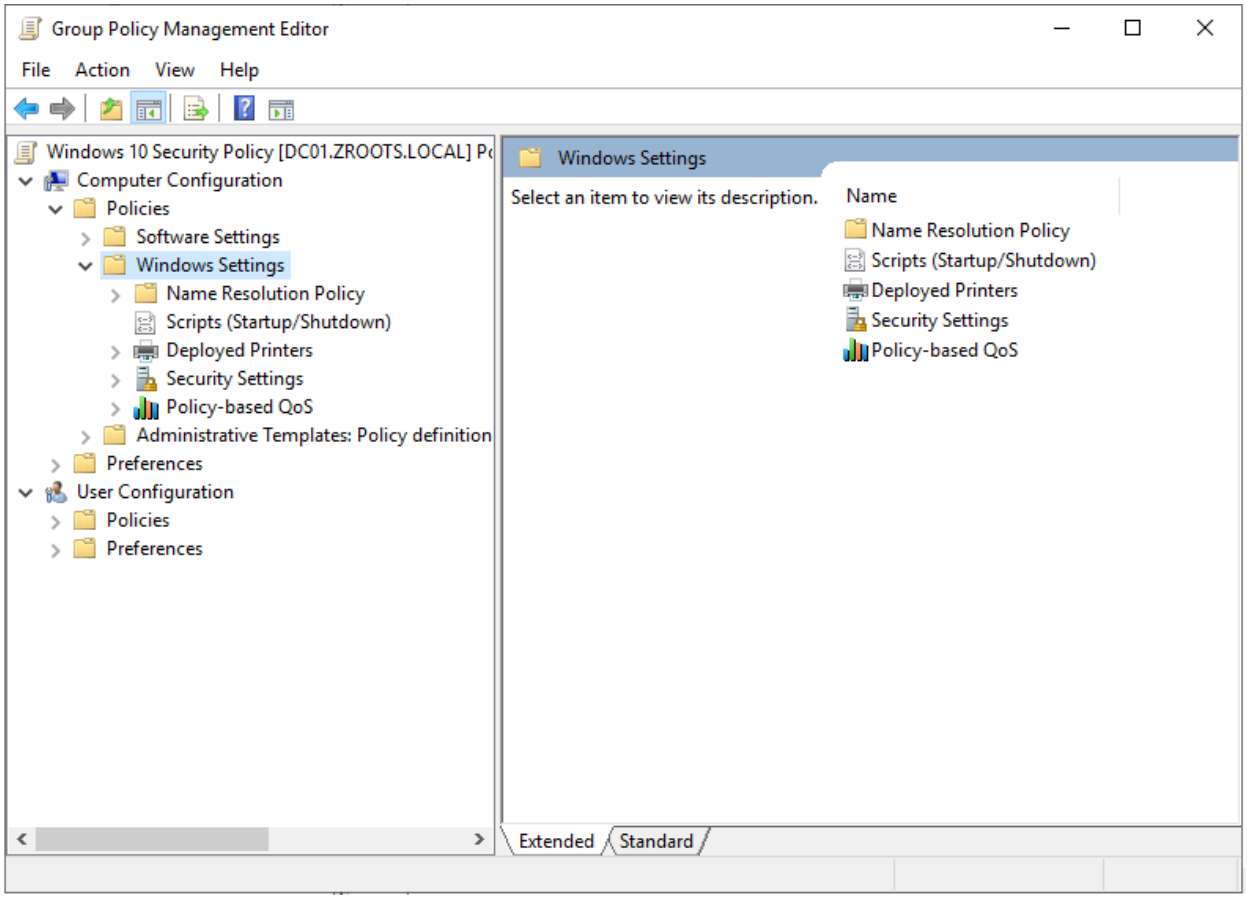
What it shows: *The Group Policy Management Editor opened on Windows 10 Security Policy [DC01.ZROOTS.LOCAL], showing the Computer Configuration and User Configuration roots.*

Figure B.13 Group Policy Management Editor Computer and User Configuration tree



What it shows: *The Policies node expanded, revealing Software Settings, Windows Settings and Administrative Templates.*

Figure B.14 Group Policy Management Editor Windows Settings and Security Settings navigation



What it shows: *Windows Settings expanded to show Name Resolution Policy, Scripts, Deployed Printers, Security Settings and Policy-based QoS, the path used to inspect security categories.*

> **Note on an omitted image**
>
> The original report included a first image of a Python/pandas traceback (a KeyError from a malware-hash scanner script). It does not show any Group Policy content and is unrelated to this lab, so it has been left out of the figures. Add it back in a separate “other session notes” section if it belongs to your records.

Appendix C Glossary

Term	Definition
Active Directory (AD)	Microsoft's directory service that stores users, computers and policy objects for a domain.
Domain Controller (DC)	A server that hosts AD and authenticates logons; DC01 in this lab.
Organizational Unit (OU)	A container in AD used to group objects and to link GPOs.
GPO	Group Policy Object a bundle of configuration settings.
Link	The association between a GPO and a site, domain or OU.
Enforced	A link option that stops lower-level GPOs from overriding the GPO's settings.
Security filtering	Limiting a GPO to specific users, computers or groups.
WMI filter	A condition (for example an OS version) that must be true for a GPO to apply.
RSoP	Resultant Set of Policy the combined effective policy after all GPOs are processed.
SYSVOL	Shared folder on DCs that stores GPO files and logon scripts.
SACL	System Access Control List defines which access attempts on an object are audited.
Fine-Grained Password Policy	A way to apply different password and lockout rules to specific users or groups in a domain.
Kerberos	The default authentication protocol in AD domains.
LSDOU	Local, Site, Domain, OU the order in which GPOs are applied.